

# Deriving the diminishing effect<sup>\*</sup>

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**Abstract.** The **diminishing effect**, a kind of norm-sensitivity, is observed in many natural language constructions: **negative comparatives** (e.g., *He is no taller than I am*), **only-sentences** (e.g., *he is only 49 years old*), and **discourse-softening markers** (e.g., Korean *cina-ci anh-* and Chinese *bú guò* ‘not beyond, nothing but, merely’), etc. All of these constructions give rise to the inference of ‘being below an expected norm’. I focus on the construction ‘**no + comparative**’ and propose that the diminishing effect arises from the interaction of two elements: (i) negation and (ii) universal quantification over items exceeding a certain degree  $d$  along a scale. I argue that (i) negating all values above  $d$  amounts to an *even*-sentence, (ii) when interpreting an *even*-sentence, we accommodate a degree QUD, such that the prejacent of *even* provides a maximally informative answer to this degree QUD, and (iii) a maximally informative degree is a superlative value with regard to a certain comparison class and is therefore necessarily norm-sensitive (see also Krifka 2000). I further extend this analysis to the diminishing effect of *only*.

**Keywords:** Diminishing effect · Norm-sensitivity · Scalarity · *Even* · *Only* · Comparatives · QUD (Question under discussion) · Informativeness · Degree QUD · NPI (Negative polarity items) · Chinese *dōu*

## 1 Introduction

The **diminishing effect** is observed in many natural language constructions, including **negative comparatives** like (1), **only-sentences** (see (2)), and **discourse-softening markers** (e.g., Korean *cina-ci anh-* and Chinese *bú guò* ‘not beyond, nothing but, merely’, see (3) and (4)).

- (1) He is **no taller than** I am. (Jespersen 1954, Nouwen 2008, Homer 2019)  
     $\rightsquigarrow$  He and I are short, i.e., our heights are below an expected norm.
- (2) He is **only** 49 years old. (see also e.g., Alxatib 2020, Greenberg 2022)  
     $\rightsquigarrow$  His age is below an expected norm.

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- (3) nà **bú-guò** shì yǔ shēng  
 that not-beyond COPULA rain sound  
 ‘That is just the sound of rain.’ **Chinese** *bú-guò* ‘merely, nothing but’  
 $\rightsquigarrow$  the sound of rain is below a threshold along a scale (e.g., importance)
- (4) ku soli-nun pi-ey **cina-ci** **anh**-a-yo.  
 that sound-TOP rain-DAT beyond-NOMINALIZER NEG-SFP  
 ‘That sound is just rain.’ **Korean** *cina-ci anh-* ‘merely, nothing but’  
 $\rightsquigarrow$  the sound of rain is below a threshold along a scale (e.g., importance)

As illustrated in (1)–(4), all these constructions (highlighted in bold) convey the inference of ‘being below a contextual norm or threshold along a scale’. In other words, these constructions demonstrate a kind of **norm-sensitivity** (or **evaluativity**), reminiscent of the **positive** use of gradable adjectives (see (5)).

However, the diminishing effects in (1)–(4) and the norm-sensitivity of the positive use of gradable adjectives do not pattern exactly the same. The norm-sensitivity of the positive use remains under negation (see (5)), while the diminishing effect of *no taller* or *only* disappears under negation (see (6)–(7)).

- (5) a. Mary is tall. Mary’s height **reaches** the threshold of being tall  
 b. Mary is not tall. Mary’s height is **below** the threshold of being tall
- (6) a. He is no taller than I am.  $\rightsquigarrow$  We are short (✓ norm-sensitive)  
 b. He is taller than I am.  $\nrightarrow$  We aren’t short (# norm-sensitive)
- (7) a. He is only 49 years old.  $\rightsquigarrow$  He is young / not old. (✓ norm-sensitive)  
 b. He isn’t only 49 years old.  $\nrightarrow$  He is not young (# norm-sensitive)

The contrast between (6a) and (6b) and that between (7a) and (7b), i.e., the presence vs. absence of norm-sensitivity, suggest that the diminishing inference of *no taller than* / *only* is more likely an implicature than a presupposition. The current paper aims to derive this implied diminishing inference.

I start with an empirical comparison between two kinds of negative comparatives: ‘*not* + comparative’ vs. ‘*no* + comparative’ (§2). Building on existing theories of comparatives (§3), I show the key difference between ‘*not* + comparative’ and ‘*no* + comparative’: the latter (i.e., *no*) includes not only **negation**, but also **universal quantification**. Thus I propose that the diminishing effect arises from the interaction between two elements: (i) negation, and (ii) universal quantification over values exceeding a certain degree *d* along a scale (§4).

In a nutshell, I argue that (i) negating all values above *d* amounts to an *even*-sentence, (ii) when interpreting an *even*-sentence, we accommodate a degree QUD (Question Under Discussion), such that the prejacent of *even* provides a maximally informative answer to this degree QUD, and (iii) a maximally informative degree is a superlative value with respect to a certain comparison class and is therefore necessarily norm-sensitive (see also Krifka 2000).

§5 extends the discussion to *only*-sentences, which are analyzed in §6. §7 addresses broader implications of the current analysis, especially its connections to NPI interpretation and maximal informativeness. §8 concludes.

## 2 Empirical observations (I): Negative comparatives

The observation of a subtle meaning difference between *not more than* and *no more than* can date back to Jespersen (1954). In his discussion of sentences (8) (from Gibbon) and (9), Jespersen (1954) notes that only (9), but not (8), ‘would have implied that the author thought this a brief period’ (p. 435, 16.844). In other words, only (9), but not (8), conveys norm-sensitivity: ‘three months’ is considered short. Similarly, Jespersen (1954) compares (10) and (11) and points out that only the latter means that both the gospel and what I tell you are lies.

- (8) The victorious emperor remained at Rome not more than three months.  
Jespersen (1954): *not more than three* = ‘three at most’  $\nrightarrow$  short
- (9) The victorious emperor remained at Rome no more than three months.  
Jespersen (1954): *no more than three* = ‘three only’  $\rightsquigarrow$  short
- (10) The gospel is not more true than what I tell you.  $\rightsquigarrow$  both can be true
- (11) The gospel is no more true than what I tell you.  $\rightsquigarrow$  **both are lies!**

More generally, ‘*not* + comparative’ is not norm-sensitive, while ‘*no* + comparative’ is norm-sensitive (see also e.g., Sawada 2005, Nouwen 2008).

In (12) and (13), both *not taller than* and *no taller than* convey the meaning that Mary’s height does not exceed Lucy’s.

- (12) Mary is not taller than Lucy, ( $\checkmark$  but both are actually tall).  
 $\rightsquigarrow$  **There is no diminishing effect**
- (13) Mary is no taller than Lucy, ( $\#$  but both are actually tall).  
 $\rightsquigarrow$  **There is a diminishing effect**: Mary and Lucy are considered short, which is at odds with the continuation here

The negative comparative in (12), *not taller than*, can naturally be followed by *but both are actually tall*, suggesting that *not taller than* is **not norm-sensitive** (or **non-evaluative**) and still compatible with a positive use of *tall*.

In contrast, adding the the same continuation *but both are actually tall* to the negative comparative in (13) (i.e., *no taller than*) results in infelicity, suggesting that the interpretation of *no taller than* is **norm-sensitive** (or **evaluative**) and at odds with a positive use of *tall* in the same context.

This continuation test thus confirms that in (13), *no taller than* brings the inference that Mary and Lucy are not considered tall, i.e., a diminishing effect.

Another piece of empirical evidence comes from the contrast between (14a) and (14b): when uttered out of the blue, only the latter sounds funny. Given that *no taller than* has a diminishing effect, (14b) suggests that 6 feet is considered short, which is at odds with our world knowledge about females’ heights.

- (14) a. Mary is not taller than 6 feet.  $\rightsquigarrow$  it sounds neutral
- b. Mary is no taller than 6 feet.  $\rightsquigarrow$  it sounds funny: is 6 feet short?!

In addition to the availability of the diminishing effect, there is also a **distributational difference** between ‘*not* + comparative’ and ‘*no* + comparative’.

As illustrated in (15) and (16), *not taller than* is compatible with a numerical differential (see *1 inch* in (15)), while *no taller than* is incompatible with such a numerical differential (see (16), which is ungrammatical).<sup>1</sup>

- (15) Mary is not 1 inch taller than Lucy.  
***Not taller than: compatible with a numerical differential***
- (16) \*Mary is no 1 inch taller than Lucy.  
***No taller than: incompatible with a numerical differential***

### 3 The analysis of comparatives

This section focuses on the case of ‘*not* + comparative’. I show that its (i) lack of norm-sensitivity and (ii) compatibility with a numerical differential can be naturally explained within our existing understanding of comparatives.

I combine (i) degree semantics (e.g., von Stechow 1984, Beck 2012) along with (ii) Zhang and Ling (2021)’s additive view of English comparative morpheme *-er* (see also Greenberg 2010) in analyzing the semantics of comparatives.

Within degree semantics, a gradable adjective like *tall* relates a degree  $d$  and an individual  $x$  (see (17)), meaning that the measurement of the individual  $x$  reaches the degree  $d$  along the scale of height, i.e.,  $x$  is tall to the extent of  $d$ .

- (17)  $\llbracket \text{tall} \rrbracket \stackrel{\text{def}}{=} \lambda d. \lambda x. \text{HEIGHT}(x) \geq d$  i.e., the height of  $x$  reaches degree  $d$

Various uses of gradable adjective *tall* can be immediately accounted for.

In the positive use of *tall* (see (18)), the degree argument of  $\llbracket \text{tall} \rrbracket$  is provided by a covert, free degree variable:  $d_{\text{pos}}$ , which represents a contextually relevant threshold of being tall. Obviously, since the positive use necessarily involves a comparison between the target (here Mary’s height) and a contextual threshold, the positive use is norm-sensitive (i.e., evaluative).<sup>2</sup>

- (18)  $\llbracket \text{Mary is POS tall} \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) \geq d_{\text{pos}}$  **Positive use**

In a measurement sentence (see (19)), the degree argument of  $\llbracket \text{tall} \rrbracket$  is provided by the measure phrase: here *6 feet*. Thus (19) means that Mary’s height reaches the degree position corresponding to 6 feet along the scale of height.<sup>3</sup>

<sup>1</sup> In addition to the availability of diminishing effects and compatibility with a numerical differential, ‘*not* + comparative’ and ‘*no* + comparative’ also differ in their prosody: English native speakers pronounce the latter as a single prosodic group.

<sup>2</sup> Actually, this degree argument in the positive use, i.e.,  $d_{\text{pos}}$  in (18), is never overtly expressed in any human language. Therefore, to better capture the compositionality and our intuitive inferences associated with the positive use, we likely need a formalization distinct from this over-simplified analysis shown in (18). But this issue is largely orthogonal to the current project. See e.g., Rett (2015), Lassiter and Goodman (2017) for recent pragmatic approaches to the positive use.

<sup>3</sup> Zhang and Zhang (2025) propose a different view: in the measure sentence (19), the measure phrase *6 feet* denotes the distance between the target (Mary’s height) and the reference position – the absolute zero point, rather than the position of ‘6 feet’.

$$(19) \quad \llbracket \text{Mary is 6 feet tall} \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) \geq 6' \quad \textbf{Measurement}$$

Then, within a categorial approach to questions (see Hausser and Zaefferer 1978; see Krifka 2011 for a review), a degree question like (20) involves lambda abstraction over the degree variable and denotes a function. The application of a MAX operator (see (21)) to this function yields the maximal degree that Mary's height reaches, i.e., Mary's height.<sup>4</sup>

$$(20) \quad \llbracket \text{How tall is Mary?} \rrbracket \Leftrightarrow \lambda d. [\text{HEIGHT}(\text{Mary}) \geq d] \quad \textbf{Degree question}$$

$$(21) \quad \text{MAX} \stackrel{\text{def}}{=} \lambda D_{\langle dt \rangle}. \iota d [d \in D \wedge \forall d' [d' \in D \rightarrow d' \leq d]]$$

As illustrated in (22), in a comparative, its *than*-clause contains an elided gradable adjective (here *tall*).<sup>5</sup> With degree abstraction and the application of a maximality operator (see (23a)–(23b)), the *than*-clause denotes the maximal degree that Lucy's height reaches, i.e., HEIGHT(Lucy).

$$(22) \quad \text{Mary is (one inch) taller than Lucy is } \text{tall}.$$

LF: Mary is tall [(one inch) -er [than [  $\lambda d'. \text{Lucy is } (d' \text{-tall})$  ] ] ]

$$(23) \quad \text{The semantics of the } \textit{than}\text{-clause of (22)}$$

- a.  $\llbracket \lambda d'. \text{Lucy is } d' \text{-tall} \rrbracket = \lambda d'. [\text{HEIGHT}(\text{Lucy}) \geq d']$   
(i.e., the set of height degrees that Lucy's height reaches)
- b.  $\llbracket \text{than} \rrbracket \stackrel{\text{def}}{=} \text{MAX} \stackrel{\text{def}}{=} \lambda P_{\langle dt \rangle}. \iota d [d \in P \wedge \forall d' [d' \in P \rightarrow d' \leq d]]$   
 $\llbracket \text{than} \rrbracket \llbracket \lambda d'. \text{Lucy is } d' \text{-tall} \rrbracket = \text{HEIGHT}(\text{Lucy})$   
 (i.e., the maximal height degree that Lucy's height reaches)

Under the canonical view (see e.g., von Stechow 1984, Beck 2012), comparative morpheme *-er/more* relates two degrees, expressing an inequality (see (24)). However, I will follow Zhang and Ling (2021)'s **additive view of -er/more**: it is an additive particle in the domain of degrees, denoting an increase along a scale (see (25)). As an additive particle, *-er/more* presupposes the existence of a salient base value for the increase. Evidence of this view comes from the parallelism between *more* and (*an*)*other* in expressing addition (see (26)).

$$(24) \quad \llbracket \text{-er} \rrbracket_{\langle d, \langle dt \rangle \rangle} \stackrel{\text{def}}{=} \lambda d_2. \lambda d_1. d_1 > d_2 \quad \textbf{Canonical view (see Beck 2012)}$$

$$(25) \quad \llbracket \text{-er} \rrbracket_d \stackrel{\text{def}}{=} d_{\text{INCREASE}} \quad \textbf{The additive view (Zhang and Ling 2021)}$$

$d_{\text{INCREASE}}$  is an unspecified positive degree (i.e.,  $d_{\text{INCREASE}} > 0$ ) that denotes an increase from a lower towards a higher position along a scale.

<sup>4</sup> Applying this MAX operator to the function representing a degree question (e.g., (20)) amounts to the application of a Dayal-style answerhood operator (Dayal 1996), yielding the maximally informative true answer to a *wh*-question. Thus, the meaning of a *than*-clause (see (23b)) amounts to the fragment answer to a corresponding degree question (e.g., *how tall is Lucy*). See Fleisher (2018), Zhang and Ling (2021).

<sup>5</sup> The evidence comes from subcomparatives like *The bathtub is wider than the door is tall*. The assumption of an elided gradable adjective in the *than*-clause (Bresnan 1973, Chomsky 1977) only works for clausal comparatives, not for phrasal comparatives. In this paper, I do not address the distinction between clausal and phrasal comparatives.

Thus, (27) shows the degree argument of  $\llbracket \text{tall} \rrbracket$  at the matrix level of sentence (22): it is a higher position along the height scale, computed from the base value (i.e., the *than*-clause) plus an increase (expressed by *(one inch) -er*).

(27) 
$$\left[ \left[ \underbrace{\text{[(one inch) -er]}_{\text{an increase}} \quad \underbrace{[\text{than } [\lambda d'. \text{Lucy is } (d' - \text{tall})]}_{\text{the base part (i.e., a lower position along the height scale)}}] \right] \right]$$

a. = HEIGHT(Lucy) +  $d_{\text{INCREASE}}$       **Without a numerical restriction**  
b. = HEIGHT(Lucy) + 1"       $d_{\text{INCREASE}}$  **is specified by 1"**

(28) a.  $\llbracket (22) \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + d_{\text{INCREASE}}$   
 $\Leftrightarrow \text{HEIGHT}(\text{Mary}) > \text{HEIGHT}(\text{Lucy})$  **Without one inch**  
 b.  $\llbracket (22) \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + 1''$  **With one inch**

(29)  $\llbracket \text{not} \rrbracket_{\langle \langle st \rangle, st \rangle} \stackrel{\text{def}}{=} \lambda p_{\langle st \rangle} . \{ w \mid p(w) = 0 \}$

(30) Mary is not (one inch) taller than Lucy is tall. **Negating (22)**  
 LF: not [ Mary is tall [(one inch) -er [than [  $\lambda d'$ . Lucy is ( $d'$  -tall) ] ] ] ]

a.  $\llbracket (30) \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) \leq \text{HEIGHT}(\text{Lucy})$  cf. (28a)

b.  $\llbracket (30) \rrbracket \Leftrightarrow \text{HEIGHT}(\text{Mary}) < \text{HEIGHT}(\text{Lucy}) + 1''$  cf. (28b)

The compatibility of ‘*not* + comparative’ with a numerical differential is also explained. In a comparative (see the LF in (22) and (30)), a numerical differential (e.g., *one inch*) restricts and specifies the default increase denoted by *-er*, just

like in (26), where the increase denoted by *more books* is restricted by *two* (see (26a)), and the increase denoted by *another book* is specified by *Different seasons* (see (26b)). In a positive comparative, adding a numerical differential restricts the size of an increase, making the comparative more informative (e.g., (28b) entails (28a)). In ‘*not* + comparative’, including a numerical differential makes the comparative less informative (e.g., (30a) entails (30b)).

Compared to the canonical view of comparative morpheme *-er/more* (see (24)), the additive view adopted here has two advantages that the canonical view can account for only with additional stipulations.

First, it provides a natural connection between the optional numerical differential and comparative morpheme *-er/more* itself. In (28)/(30), without a numerical differential, the increase  $d_{\text{INCREASE}} \in (0, +\infty)$  (i.e., the lower bound is open), whereas with a numerical differential, the increase becomes more restricted:  $d_{\text{INCREASE}} \in [1'', +\infty)$  (i.e., the lower bound is closed). Thus no stipulation is needed to explain how including a numerical differential changes a strict inequality into a non-strict inequality in (28).

Second, as a consequence of the first advantage, the current analysis naturally explains the different patterns between positive and negative comparatives regarding the strictness of inequalities. In positive comparatives (see (28)), the comparison is a strict inequality when there is no numerical differential, (see (28a)), but becomes a non-strict inequality when there is a numerical differential (see (28b)). In contrast, this pattern is reversed for ‘*not* + comparatives’ (see (30)): the comparison is a non-strict inequality when no numerical differential is present (see (30a)), but becomes a strict inequality when there is a numerical differential (see (30b)).<sup>6</sup>

## 4 Accounting for the diminishing effect of *no taller than*

(i) The incompatibility of ‘*no* + comparative’ with a numerical differential and (ii) its diminishing effect (i.e., the inference of an inequality in comparison with a contextual norm) are fundamentally due to the contribution of *no*. Below I focus on how the use of *no* in a comparative gives rise to the diminishing inference.

<sup>6</sup> Nouwen (2008) claims that the comparatives in (i), which express a non-strict inequality, should have an exhaustive implicature and thus a strengthened equality reading. I think that only (ia), but not (ib), has a strengthened equality reading, and that for (ia), this reading is naturally projected from the ‘exactly’ interpretation of *1 inch*. On the other hand, I will show that ‘*no* + comparative’ differs from ‘*not* + comparative’ (see (ib)) and does lead to a strengthened equality-like meaning.

- |     |    |                                                                                    |                                             |
|-----|----|------------------------------------------------------------------------------------|---------------------------------------------|
| (i) | a. | Mary is 1 inch taller than Lucy.                                                   | (=(28b))                                    |
|     |    | $\Leftrightarrow \text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + 1''$ | $d_{\text{INCREASE}} \in [1'', +\infty)$    |
|     |    | $\rightsquigarrow \text{HEIGHT}(\text{Mary}) = \text{HEIGHT}(\text{Lucy}) + 1''$   | (Implicature: $d_{\text{INCREASE}} = 1''$ ) |
|     | b. | Mary is not taller than Lucy.                                                      | (=(30a))                                    |
|     |    | $\Leftrightarrow \text{HEIGHT}(\text{Mary}) \leq \text{HEIGHT}(\text{Lucy})$       |                                             |
|     |    | $\not\rightsquigarrow \text{HEIGHT}(\text{Mary}) = \text{HEIGHT}(\text{Lucy})$     | (cf. Nouwen 2008)                           |

#### 4.1 Decomposing negative quantifier *no*

I follow the long existing compositional view on *no* (e.g., [Jacobs 1980](#)) and split it into (i) negation operator *not* and (ii) an existential quantifier (see (31)).

- (31) No students came.  $\llbracket \text{no} \rrbracket \stackrel{\text{def}}{=} \lambda P. \lambda Q. \neg \exists x [P(x) \wedge Q(x)]$   
 LF: Not [a student came]  $\llbracket (31) \rrbracket \Leftrightarrow \neg \exists x [\text{STUDENT}(x) \wedge \text{CAME}(x)]$

In *no taller than*, *no* is decomposed into (i) *not* and (ii) an existential quantifier over the domain of degree values. As illustrated in (32), this sentence means that ‘there is no positive increase of the size *d* s.t., Mary is *d*-taller than Lucy’.

- (32) Mary is no taller than Lucy is ~~tall~~.  
 LF:  $\neg \exists d. [\text{Mary is tall } [d \dots \text{-er} [ \text{than } [ \lambda d'. \text{Lucy (is } d'\text{-tall)} ] ] ] ]$   
 $\llbracket (32) \rrbracket \Leftrightarrow \neg \exists d. [\text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + d]$  (here  $d > 0$ )  
 i.e., there doesn’t exist a positive degree, *d*, such that Mary’s height reaches the position ‘HEIGHT(Lucy) + *d*’ along the height scale.

Under the current compositional analysis of *no*, the incompatibility of ‘*no* + comparative’ with a numerical differential is straightforwardly explained. A numerical differential functions as a restriction on a discourse referent (dref) that denotes an increase. In both sentences in (33), since ‘ $\neg \exists$ ’ is used to bind a variable, no dref is established for the restriction to attach to.

- (33) a. \*Mary is no 1 inch taller than Lucy is.  
 b. \*No other student, Bill, came.

In contrast, in ‘*not* + comparative’, an increase does exist as a dref, so that it can be further restricted (see the contrast between (34) and (33)).

- (34) a. Mary is not (1 inch) taller than Lucy.  
 i.e., there is a position with an increase (of one inch) above Lucy’s height, but Mary’s height does not reach that position.  
 b. Four boys came. Another boy, (Bill,) didn’t come.

#### 4.2 Unfolding universal quantification

Given that  $\neg \exists d [P(d)]$  and  $\forall d [\neg P(d)]$  are truth-conditionally equivalent under classical first-order logic, I recast a ‘*no* + comparative’ sentence from a negated existential quantification into a universal quantification:

- (35) Mary is no taller than Lucy.  
 $\Leftrightarrow \neg \exists d [\text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + d]$  (see (32))  
 $\Leftrightarrow \forall d \neg [\text{HEIGHT}(\text{Mary}) \geq \text{HEIGHT}(\text{Lucy}) + d]$   
 $\Leftrightarrow \forall d [\text{HEIGHT}(\text{Mary}) < \text{HEIGHT}(\text{Lucy}) + d]$

This universal quantification in (35) can be further unfolded as the conjunction of a set of propositions (see e.g., [Bumford 2015](#)):

- (36)  $\{\dots,$   
 $\text{HEIGHT}(\text{Mary}) < \text{HEIGHT}(\text{Lucy}) + 1'',$   
 $\text{HEIGHT}(\text{Mary}) < \text{HEIGHT}(\text{Lucy}) + 2'',$   
 $\dots\}$

Given that the domain of  $d$  (height increases) is a **totally ordered set** of degrees, the set of propositions in (36) is also a totally ordered set, along a scale of informativeness. Among these propositions in the set of (36), ‘ $\text{HEIGHT}(\text{Mary}) < \epsilon + \text{HEIGHT}(\text{Lucy})$ ’ (where  $\epsilon$  represents the minimal height increase) is the most informative proposition, entailing all others.

Thus, the meaning of the above *no-taller-than* sentence (see (35)) is equivalent to the interpretation of this maximally informative proposition in the ordered set in (36), i.e., a sentence with a minimizer-NPI (see (37)).

- (37) Mary is no taller than Lucy.  
 $\Leftrightarrow \text{HEIGHT}(\text{Mary}) < \epsilon + \text{HEIGHT}(\text{Lucy})$   
 $(\epsilon: \text{the minimal element in its alternative set})$   
 $\Leftrightarrow \text{Mary is not } \{ \text{any} / \text{a tiny bit} / \text{slightly} \} \text{ taller than Lucy}$

To account for the diminishing effect of *no taller than*, now I need to explain how this NPI-sentence *Mary is not a tiny bit taller than Lucy* (i.e.,  $\text{HEIGHT}(\text{Mary}) < \epsilon + \text{HEIGHT}(\text{Lucy})$ ) gives rise to the norm-sensitive inference that  $\text{HEIGHT}(\text{Mary})$  and  $\text{HEIGHT}(\text{Lucy})$  are below the expected norm.

### 4.3 Accommodating a hidden *even*

I propose that the sentence *Mary is not a tiny bit taller than Lucy* is interpreted in the same way as the corresponding sentence with a hidden *even*:

- (38) a. Mary is not a tiny bit taller than Lucy  
 b.  $\rightsquigarrow$  Mary is not **EVEN** a tiny bit taller than Lucy  
 $= \text{Mary is not } \text{EVEN} \text{ as tall as } [X]_F \quad (X = \text{HEIGHT}(\text{Lucy}) + \epsilon)$

This accommodation is in the same spirit of Heim (1991)’s pragmatic principle *Maximize Presupposition* (see also Schlenker 2012):

- (39) **Maximize Presupposition** (at the sentence level) (Schlenker 2012)  
 If a sentence  $S$  is a presuppositional alternative of a sentence  $S'$  (i.e.,  $S \in \text{Alt}(S')$ ) and the context  $C$  is such that  
 (i) the presuppositions of  $S$  and  $S'$  are satisfied within  $C$ ;  
 (ii)  $S$  and  $S'$  have the same assertative component relative to  $C$ ;  
 (iii)  $S$  carries a stronger presupposition than  $S'$ ,  
 then  $S$  should be preferred to  $S'$ .

Below I first present the presuppositions triggered by *even*. Then I show (i) that (38a) is already maximally informative in its alternative set (see (36)) and thus freely supports the accommodation of a hidden *even*, and (ii) that, with a hidden *even*, (38b) has a stronger presupposition and is thus preferred.

**1. The presupposition triggered by *even*** According to the traditional view (e.g., [Horn 1969](#)), the use of *even* brings two presuppositions: (i) **entity-based additivity** (see [\(40a\)](#)) and (ii) **likelihood-based scalarity** (see [\(40b\)](#)).

- (40) (It's not the case that) even [Mary]<sub>F</sub> came.  
 a. [\(40\)](#)  $\models$  Someone other than Mary came.  
 b. [\(40\)](#)  $\models$  Compared to others, Mary was unlikely to come.

However, as illustrated by an example from [Szabolcsi \(2017\)](#), under the scenario in [\(41\)](#), the use of the *even*-sentence in [\(42\)](#) is perfectly natural, yet it challenges the traditional view. First, the entity-based additivity requirement is not satisfied (see [\(42a\)](#)), because Eeyore is the only one who took a bite of the thistles and spit them out. Second, if no one other than Eeyore spit out the thistles, then it is unclear how the prejacent could be less likely to be true than alternatives like *X spit the thistles out* ( $X \in \text{Alt}(\text{Eeyore})$ , see [\(42b\)](#)).

- (41) **Scenario:** Imagine Pooh and friends coming upon a bush of thistles. Eeyore (known to favor thistles) takes a bite but spits it out.  
 (42) (Those thistles must be really prickly!) Even [Eeyore]<sub>F</sub> spit them out!  
 a. [\(42\)](#)  $\not\models$  Someone other than Eeyore spit thistles out.  
 b. [\(42\)](#)  $\not\models$  Compared to others, Eeyore was unlikely to spit thistles out.

[Zhang \(2022\)](#) proposes a degree-QUD-based analysis of the use of *even* (see [Greenberg 2018](#) for a similar view). An *even*-sentence asserts its prejacent and has the presuppositions shown in [\(43\)](#):

- (43) The presuppositions of *even*(*p*) ([Zhang 2022](#), see also [Greenberg 2018](#)):  
 a. There is a contextually salient degree QUD  
 b. *p* is maximally informative in resolving the degree QUD  
     ( $\rightsquigarrow$  Consequently, *p* solves the degree QUD with norm-sensitivity)

This degree-QUD-based view explains the potential lack of entity-based additivity or low likelihood in interpreting an *even*-sentence. Under the scenario in [\(41\)](#), no one other than Eeyore took a bite of the thistles. Thus the felicitous use of *even* in [\(42\)](#) does not require its prejacent to be less likely than alternatives.

Rather than addressing who spit the thistles out or who was the least likely to do so, [\(42\)](#) addresses a degree QUD: *how prickly the thistles are*. Compared to its alternatives (e.g., *Pooh spit them out*), the prejacent *Eeyore spit them out* is maximally informative in addressing this degree QUD.

**2. The accommodation of a hidden *even*** Given the degree-QUD-based analysis of *even*, a sentence containing a minimizer-NPI naturally satisfies the presuppositional requirements of *even*, provided that the discourse supplies a relevant, salient degree QUD. As illustrated in [\(44\)](#), when the QUD is about the degree of John's mental withdrawal, sentence [\(44a\)](#) can be interpreted as if it contained a covert *even*, yielding a pragmatically strengthened interpretation.

- (44) Context: John was mentally withdrawn and closed off in his own mind.
- a. The party was crowded, but John didn't see a single person.
  - b. The party was crowded, but John didn't EVEN see a single person.

Without a covert *even*, (44a) simply means that John saw no one. The use of the NPI, *a single person*, already makes (44a) the most informative member of its alternative set (e.g., *John didn't see a single person* entails *John didn't see two people, three people*, etc.). With a covert *even*, (44b) further means that, compared to its alternatives (e.g., *John didn't see two people*), the prejacent of *even*, *John didn't see a single person*, is maximally informative in resolving a salient degree QUD: e.g., here *how mentally withdrawn / closed off John is*.

A sentence like (45a) can also be interpreted as if it contained a covert *even*. With the NPI *a tiny bit*, (45a) is the most informative among its alternatives. Therefore, (45a) naturally meets the presuppositional requirements of *even*, provided that there is a salient degree QUD. Since the sentence already has a gradable adjective, a salient degree QUD is easy to accommodate.

- (45) a. Mary is not a tiny bit taller than Lucy.  
 b. Mary is not EVEN a tiny bit taller than Lucy.  
     = Mary is not EVEN as tall as  $[X]_F$       ( $X = \text{HEIGHT}(\text{Lucy}) + \epsilon$ )

With a hidden *even*, the prejacent in (45b) become closely connected to a degree QUD and resolve it with maximal informativeness. Since *a tiny bit* is a minimizer,  $X$  is the lowest value in its alternative set. A felicitous degree QUD must be such that, in resolving it, a lower height value corresponds to a higher level of informativeness: e.g., *how short is Mary?*, *how not tall is Mary?*. As I will show below, the accommodation of such a degree QUD brings a further pragmatic consequence: the prejacent of *even* becomes necessarily norm-sensitive.<sup>7</sup>

<sup>7</sup> A reviewer raises two questions regarding (45). First, is there any meaning difference between (45a) and (45b)? In my view, there is almost no difference.

Under the current degree-QUD-based theory, *even* (i) invokes a salient degree QUD and (ii) requires its prejacent to be maximally informative in resolving this degree QUD. In (45), regardless of whether a covert *even* is assumed, the NPI-sentence (45a) is already maximally informative in its alternative set.

Moreover, given the presence of a gradable adjective (cf. (44)), this comparative sentence (45a) already naturally addresses a degree-related issue, without relying on a covert *even*. In other words, positing a hidden *even* brings no additional meaning beyond what is already conveyed by *Mary is not a tiny bit taller than Lucy*.

Nevertheless, positing a hidden *even* pragmatically strengthens the connection between this sentence and a degree QUD. As I will show below, it is maximal informativeness in addressing such a degree QUD that gives rise to norm-sensitivity.

The reviewer also asks whether the norm-sensitive inference hinges on which specific degree QUD is accommodated. As addressed in the main text, in interpreting (45), 'a felicitous degree QUD must be such that, in resolving it, a lower height value corresponds to a higher level of informativeness'. Thus, degree QUDs like *how short is Mary?* or *how not tall is Mary?* all meet this requirement and yield the same norm-sensitive inference. See also Zhang (2025) for further discussion.

Before proceeding, I would like to briefly clarify two issues.

First, does the principle of *Maximize Presupposition* allow a strengthened interpretation that involves a hidden presupposition trigger? My answer is *yes*. As explained by Schlenker (2012), a strengthened meaning based on *Maximize Presupposition* can be considered a kind of scalar implicature. In (46), unique existence is more informative than mere existence, which results in interlocutors' preference for (46b) when the uniqueness requirement is met.

- (46) a. #A sun is shining. the existence of a sun  
b. The sun is shining. the unique existence of a sun

However, the principle of *Maximize Presupposition* does not mean that in natural language, whenever the presuppositional requirements of a presupposition trigger are met, the presupposition trigger has to be overtly pronounced. In (47), the additivity requirement of *too* is certainly met, but the overt presence of *too* is not required, but optional.

- (47) A<sup>u</sup> girl came. Then another<sup>v</sup> girl came, (too).

Presumably, different linguistic constructions can overlap in contributing to enhancing informativeness. In (47), the use of *another* already contributes the meaning of additivity. As a result, the hearer can arrive at a sufficiently informative interpretation without relying entirely on the use of *too* (which further brings the meaning of parallelism), so the overt presence of *too* is optional.

In (44) and (45), the minimizers *a single person* and *a tiny bit (taller)* already introduce an ordered set of alternatives and thus establish an informativeness ordering. The use of *even* further exploits this ordering in addressing a degree QUD. In other words, the informativeness ordering and maximal informativeness exist independently of *even*. Thus the overt presence of *even* is optional.

By contrast, in (42), *Even [Eeyore]<sub>F</sub> spit the thistles out*, the focused part *Eeyore*, unlike an NPI, cannot by itself introduce an ordered set of alternatives. Instead, it is the semantics of *even* that invokes the relevant degree QUD (i.e., *how prickly the thistles are*), thereby imposing an ordering: Eeyore is (one of) the most thistle-tolerant animal(s) in its set of alternatives. Thus, in this case, the ordering of informativeness (and hence maximal informativeness) entirely hinges on *even*, making its presence obligatory.<sup>8</sup>

Another issue, closely related to the first, concerns the potential issue of over-generation: can the accommodation of a covert *even* lead to unattested strengthened interpretations? My answer is *no*.<sup>9</sup> For example, the interpretation of (48a) cannot involve a covert *even* in a way that would strengthen its meaning to that of (48b). In (48a), *[60 kg]<sub>F</sub>* does not by itself invoke an ordered set of alternatives that would associate the value of *60 kg* with maximal informativeness, and therefore cannot support the accommodation of a covert *even*.

<sup>8</sup> This is just a sketch showing that *Maximize Presupposition* is compatible with a hidden presupposition trigger. I thank Zhuang Chen for valuable discussions. Formal details are left for future work.

<sup>9</sup> I thank an anonymous reviewer for raising this issue.

Besides, the interpretation of the *even*-sentence in (48b) also differs from the cases in (44) or (45): in (48b),  $[60 \text{ kg}]_F$  is not necessarily the minimal value in its alternative set. For instance, if John aims to lose weight, (48b) would mean that he is still too heavy. Therefore, (48b) does not necessarily have the diminishing inference that his weight falls below a contextual norm.

- (48) a. John's weight is not  $[60 \text{ kg}]_F$ .  
 b. John's weight is not even  $[60 \text{ kg}]_F$ .

#### 4.4 From maximal informativeness to norm-sensitivity

Given the presuppositional requirements of *even* (see (43)), its prejacent resolves a contextually salient degree QUD with maximal informativeness. Thus, as shown in (49), the prejacent of *even* resolves a degree QUD such as *how short Mary is* with maximal informativeness. In other words, with a covert *even*, (49) means that Mary's height is associated with a superlative value, i.e., Mary is among the shortest (or the least tall). Thus Mary is short (or not tall).

- (49) Mary is not EVEN a tiny bit taller than Lucy.  
 = Mary is not EVEN as tall as  $[X]_F$  ( $X = \text{HEIGHT}(\text{Lucy}) + \epsilon$ )  
 (An appropriate degree QUD: how short is Mary? how not tall is Mary?)

There are two critical points in the current analysis. First, a superlative value is necessarily norm-sensitive. In his analysis of *already*- and *still*-like particles, Krifka (2000) explains the connection between a superlative value and a contextual norm (e.g., the threshold of being short for a relevant comparison class):

'If we assume that the 'expected' age is the average of the alternatives, then we can reconstruct one typical interpretation of sentences with *schon* and *erst*, namely that they express a deviation from expected values in a particular direction: *Schon* in *Lydia ist schon [drei]<sub>F</sub> Monate alt* 'Lydia is already 3 months old' gives rise to the understanding that Lydia's age is greater than may have been expected, and *erst* in *Lydia ist erst [drei]<sub>F</sub> Monate alt* 'Lydia is only 3 months old', that it is smaller than may have been expected.' (Krifka 2000: p. 405)

In *Mary is not EVEN a tiny bit taller than Lucy* (see (49)), a *tiny bit* ( $= \epsilon$ ) is a minimizer, so ' $\text{HEIGHT}(\text{Lucy}) + \epsilon$ ' is the lowest value among its alternatives, lower than an expected norm of tallness *Exp*. Consequently, both  $\text{HEIGHT}(\text{Lucy})$  and  $\text{HEIGHT}(\text{Mary})$  are below *Exp*, yielding the norm-sensitive inference.

Second, since the prejacent of *even* is maximally informative in addressing a salient degree QUD, there is no higher level of informativeness. For example, in (50), the focus associate of *even*, 3, is maximally informative in addressing the relevant degree QUD. Using a higher number (e.g., 4) would not increase informativeness: the burden is still interpreted as maximally heavy.

- (50) (QUD: *how heavy is the burden of parenting?*) Mary even has  $[3]_F$  kids.

Similarly, in interpreting *Mary is not EVEN a tiny bit taller than Lucy*, any heights not as tall as ‘HEIGHT(Lucy) +  $\epsilon$ ’ count as maximally informative, and all such values are considered among the shortest. This explains the intuitive equality-like inference for *Mary is no taller than Lucy*: Mary and Lucy are equally among the shortest.

To sum up, ‘*no* + comparative’ expresses the negation of all values above a certain degree  $d$ . This yields a totally ordered set of propositions along the scale of informativeness and is pragmatically strengthened to an *even*-sentence. The maximal informativeness of the preajcent of *even* leads to norm-sensitivity.

## 5 Empirical observations (II): *Only*-sentences

Similar to the construction of ‘*no* + comparative’, *only*-sentences also exhibit a diminishing effect. In (51a), the presence of *only* makes the continuation *which is quite old* sound unnatural, because *only* triggers the inference that Bill is not old. By contrast, without *only*, (51b) naturally allows this continuation. Similar to *Mary is no taller than 6 feet*, (52) sounds funny when uttered out of the blue: the diminishing inference conflicts with our world knowledge about human age.

- (51) a. Bill is only 29, (# which is quite old). ✓ Diminishing effect  
 b. Bill is 29, (✓ which is quite old). # Diminishing effect
- (52) Kim is only 109.  $\rightsquigarrow$  it sounds funny: maybe Kim is a young vampire

## 6 Accounting for the diminishing effect of *only*-sentences

According to the canonical view (see e.g., Horn 1969, among others), an *only*-sentence has a positive inference and a negative inference (see (53)). The positive inference is often analyzed as a presupposition (see e.g., Horn 1969, Coppock and Beaver 2014) or an implicature (see e.g., van Rooij and Schulz 2007, Ippolito 2007), while the negative inference is the asserted meaning of an *only*-sentence.

- (53) Bill is only 29. ✓ **Diminishing effect**  
 a. Positive inference: Bill is 29.  
 b. Negative inference:  
   (i) #Bill is not more than 29. **No diminishing effect**  
   (ii) Bill is no more than 29. ✓ **Diminishing effect**

My discussion above on ‘*not* + comparative’ and ‘*no* + comparative’ suggests that the negative inference of an *only*-sentence should be analyzed as a ‘*no* + comparative’ construction, rather than a ‘*not* + comparative’ construction.

Accordingly, I propose to decompose *only* into (i) negative quantifier *no* and (ii) an additive / comparative component *-er/more*. *Only* expresses the negation of all values above a certain value along a scale. Based on this, as shown in (54), the diminishing inference that Bill is young (or Bill is not old) follows naturally.

- (54) Bill is only 29.  
 $\Leftrightarrow$  Bill is no more than 29.  
 $\approx$  Bill is not EVEN a tiny bit older than 29  
 $=$  Bill is not EVEN as old as  $[X]_F$  ( $X = 29 + \epsilon$ )  
 $\rightsquigarrow$  The prejacent of the hidden *even* is maximally informative in addressing a degree QUD such as *how young is Bill*, *how not old is Bill*  
 $\therefore$  Bill is young / not old

The current analysis naturally extends to other uses of *only*. As shown in (55) and (56), an *only*-sentence is analyzed as a ‘no + comparative / additive’ construction and pragmatically strengthened into an *even*-sentence. The strengthened interpretation requires the accommodation of a salient degree QUD (e.g., the degree of prominence, the size of the guest group), and maximal informativeness results in the diminishing effect. Thus, the interpretation of an *only*-sentence always involves a hidden *even*, which explains why we intuitively feel that *only* and *even* constitute a pair of scalar anatomy (see Greenberg 2022).

- (55) John is only a graduate student. Scalar use of *only*  
 $\Leftrightarrow$  John is no more than a graduate student.  
 $\approx$  John is not EVEN slightly more prominent than a graduate student.  
 $\therefore$  John is not considered prominent.
- (56) Mary only invited John and Kim.  
 $\Leftrightarrow$  Mary invited no one other than John and Kim.  
 $\approx$  Mary didn’t EVEN invite a single person other than John and Kim.  
 $\therefore$  Mary invited few people. / Mary didn’t invite many people.

It is worth noting that under the current compositional analysis, the diminishing effect of *only* comes from *only* itself (which amounts to a ‘no + comparative’ construction), rather than from the focus associate of *only*. Even when the focus associate of *only* denotes a large value in our world knowledge (e.g., *Kim is only 109* in (52)), *only* triggers a diminishing inference.

Thus the current analysis differs from the ‘exceptive’ view of *only* (see von Stechow 2007, Fintel and Iatridou 2007, Crnić 2024). The current analysis is compatible with the empirical fact that the focus associate of *only* is not necessarily an exception (see (57)) and explains the diminishing effect even in such cases: e.g., in (57), ‘90%’ is below the expected norm.

- (57) Only  $[90\%]_F$  of the students came to the final.  
 ‘Exceptive’ view: Only  $[90\%]_F$  of the students = no students but 90%

## 7 Discussion

### 7.1 NPI and maximal informativeness

In formal semantics, the norm-sensitivity of the positive use of gradable adjectives has been extensively investigated (see e.g., Rett 2015, Lassiter and Goodman 2017). The current study extends this line of inquiry by examining norm-

sensitivity beyond the positive use. Building on Krifka (2000), I show that maximal informativeness is an important source of norm-sensitivity. This view has broader theoretical implications for a wide range of phenomena, including NPIs and *wh*-exclamatives such as *what a day* (Zanuttini and Portner 2003).

Within the NPI literature, Lahiri (1998) uses Hindi data to show that an NPI is often composed from an *even*-like operator and a minimizer. In contrast, the current paper argues that an *even*-like operator and a minimizer can each contribute to maximal informativeness independently. Thus, an *even*-like operator is not necessarily hard-wired in the semantics of an NPI.

The Chinese data in (58) provides empirical evidence: a minimizer alone already behaves like an NPI.

- (58) tā duì nóng-yè yí qiào (dōu) bù tōng  
 3SG towards agriculture one point (EVEN) NEG understand  
 ‘He doesn’t understand anything about agriculture.’ Chinese  
 (*yí qiào* ‘one point’: a minimizer; *EVEN* can be optionally inserted)

On the other hand, the current paper also shows that an *even*-like particle can always be optionally included in a sentence containing a minimizer (see (58)). Both elements converge in expressing maximal informativeness.

The current view also explains why, cross-linguistically, an *also*-like additive particle (e.g., Chinese *yě*, Korean *to*, Japanese *mo*) often appears to have a scalar use in the presence of an NPI. The scalar interpretation is not due to this *also*-like particle, but due to the minimizer, which expresses maximal informativeness.

- (59) tā yī fēn qián yě méi yǒu  
 3SG one cent money also NEG have  
 ‘He doesn’t have one cent.’ Chinese

## 7.2 The cancelability of the norm-sensitivity of *even* and *only*

The current paper analyzes the norm-sensitivity of both *even* and *only* as pragmatic implicatures: norm-sensitivity is a by-product of degree-QUD-based maximal informativeness. This view differs from earlier works like Greenberg (2022), Zhang (2022), in which the norm-sensitivity of *even* is semantically hard-wired.

Greenberg (2022) argues that norm-sensitivity is cancelable for *only* and thus an implicature, but norm-sensitivity is non-cancelable for *even* and thus hard-wired (see (60)). However, it is likely that both are non-cancelable implicatures.

The felicitous use of *only* in (60a) does not mean that its diminishing effect disappears. The interpretations of *he is only 88* and *Well, B is also old* might involve different norms (see also (61): *already* still conveys ‘older than expected’). The contrast shown in (60) might stem from the easiness/difficulty of accommodating a degree QUD for *only* vs. *even*. This issue is for future research.

- (60) a. A is old. She is 92. B is younger. He is **only** 88. Well, B is also old.  
 b. C is young. He is 11. D is older. He is (**#even**) 12. ~~D is still young.~~  
 (61) E is young. She is 11. F is older. He is **already** 12. Well, F is still young.

### 7.3 Chinese *dōu*

The multi-functionality of Chinese *dōu* has also been heatedly studied for decades. As illustrated in (62) and (63), it functions like both English *all* and *even*.

The current work is in line with works like Liu (2017) and holds the view that maximal informativeness underlies both uses of *dōu* (see the connection between universal quantification and the maximally informative proposition in an ordered alternative set in §4). This view also explains the contrast in (64): the norm-sensitivity of *dōu* is also relevant in its *all*-like use, and, in a salient context, *the majority* is just like *the totality* in achieving maximal informativeness.

- (62) háizi-men dōu lái-le lái-le  
 kid-PL ALL/EVEN come-PFT  
 ‘All the kids came.’ *dōu*: a sum operator or a universal quantifier
- (63) [tā]<sub>F</sub> dōu lái-le lái-le  
 3SG ALL/EVEN come-PFT  
 ‘Even [he]<sub>F</sub> came.’ *dōu*: an *even*-like particle
- (64) { dà-bùfèn / \*xiǎo-bùfèn } háizi-men dōu lái-le lái-le  
 big-part small-part kid-PL ALL/EVEN come-PFT  
 ‘The majority of the kids came.’ (#The minority of the kids came.)

## 8 Conclusion

This paper derives the diminishing effect for natural language constructions like ‘*no* + comparative’ and *only*-sentences. I have shown that maximal informativeness is a key source of norm-sensitivity. This perspective not only unifies a range of empirical phenomena but also carries significant theoretical implications.

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